

## POLYAMIDE FIBRES- **NYLON** AND ARAMID

### Nylon

#### **Production and Manufacture of Nylon**

Nylon was formed as a result of research study by du Pont Company in 1927 – 28. Nylon is a generic term. Nylon is defined as a manufactured fibre in which the fibre forming substance is long-chain synthetic polymer. The polymer is large molecule formed by uniting small molecules in the process of polymerisation.

#### **Manufacture**

- Nylon 6.6 (six carbon atoms in each of the two chemicals used in Making the product) is made of basic chemicals **hexamethylene diamine** and **adipic acid**, each containing six carbon atoms per individual molecule.
- The nylon is made by linear condensation polymerisation process. This means that a molecule of the acid and a molecule of the diamine join together, small molecule (in case of these two water) is given off as a by-product of the reaction.
- Modern methods of manufacture involves the production of adipic acid from cyclohexane and hexamethylene from **benzene**. Cyclohexane is usually derived from distillation of petroleum.
- Specific amounts of the acid and the diamine are combined in solution to form a salt nylon salt called aminohexamthyle adipamide.
- This nylon salt is purified and the polymerised in an autoclave under an atmosphere of nitrogen.
- It is extruded in a ribbon form and chipped into small **flakes or pellets**.
- The polymer then is re-melted and extruded through a spinnerets into cold air where nylon filament is formed.
- After cooling the filament are stretched or cold drawn to orient the molecules in the fibre and develop fibre strength and fineness. Nylon fibres may be heated slightly to increase the ease at speed of drawing.
- Alternatively, the polymer may be produced continuously and pumped directly to the spinnerets instead of being made into flake or pellet form first.

#### **PROPERTIES OF NYLON**

#### Microscopic Properties

Cross-sectional shape: Normally round shape but the cross sectional shape could be controlled. Longitudinal shape is normally a smooth rod.

Longitudinal View of  
delustured nylon fibers  
of delustured nylon



Cross-sectional View



Source: <http://sy.zlge.org/Upload/20080530110027750.pdf>

### **Physical Properties**

**Length:** Length of nylon filament is unlimited and staple fibre lengths are controllable.

**Fineness:** Nylon fibre fineness is also controllable.

#### **Tenacity/Strength**

The strength of nylon fibre is very high. Its tenacity varies from 4.6 to 5.8 g/den.

#### **Elastic Recovery and Elongation**

Nylon is highly extensible fibers. Extension at break is 30% but the problem is poor recovery form extension. For that reason it is not used as sewing thread for garment.

#### **Resiliency**

It has good resiliency property. Nylon fibers, yarns or fabrics do not crease easily. For that reason it is widely used for the pile fabric production, for instance velvet, carpet etc.

**Frictional Resistance:** Nylon fibre shows good frictional resistance. Due to high strength and good frictional resistance property it is widely used for rope.

#### **Moisture Regain**

Moisture regain of nylon fibre varies from 4.2 to 5.0%. Nylon fibre does not absorb water easily. Water molecules cannot enter into the inside of the fibre easily. For this reason nylon fabrics can be dried easily. So it can be used as raincoat.

### **Chemical properties**

#### **Action of Bleaching Agents:**

Chlorine and strong oxidizing bleaching agents can damage nylon fibers but normal bleaching agent can't damage.

**Action of acid and alkali:** Strong alkali can partially damage the nylon fibres. Nylon can be dissolved in mineral acid. E.g., Hydrochloric acid and sulphuric acid.

**Action of Organic Solvents:** Nylon is safe in most organic solvents. Hence a nylon fiber is dry- cleanable. Spots could be removed by reagents.

### **Sunlight, Age and miscellaneous factors**

#### **Action of Light and Heat**

Long exposure to sunlight reduces strength of nylon fibers. It melts when in contact to flame and burns with low flame, extinguished itself.

#### **Dye Fastness**

Nylon fibers show good color fastness. It could be dyed with direct, acid, vat, disperse, basic and metallised dyes.

#### **Biological Properties**

It is attacked by fungus but it can't damage the fibers. It is also attacked by moths and insects but remain good condition.

#### **Summary of Properties of the nylon:**

- 1) It is strong and elastic.
- 2) It is easy to launder.
- 3) It dries quickly.
- 4) It retains its shape.
- 5) It is resilient and responsive to heat setting.

## Use care of nylon

### Care

- Machine wash in cold water.
- Wash nylon separately from other types of fabrics.
- If needed, use only non-chlorine bleach to bleach your white nylon garments.
- Turn the washing machine setting to low before tumbling dry.

Remove your nylon garments promptly from the tumbler after the wash is complete.

### Uses

Nylon is popular for hosiery and is used to make track pants, shorts, swimwear, active wear, windbreakers, bedspread and draperies.

It can also be made into parachutes, flak vests, combat uniforms, tires and life vests

It is also used as the netting of bridal veils, and for umbrellas and luggage bags.

It is widely used for knit fabric, socks, under garments

### Other Tips:

Extensive washing and drying in an automatic dryer can lead to piling.

White nylon should be washed separately to avoid it turning grey.

To avoid yellowing of your white nylon, you should bleach it frequently with Sodium Perborate bleach.

## ARAMIDE

### Definition

Aramide is defined as a manufactured fibre in which the fibre forming substance is long-chain synthetic polyamide. Linkages are attached directly to two aromatic rings.

There are different types e.g. this fibre e.g. Normex, Kevlar, conex, fenilon, Arenka and kermel. Normex and kervlar from Du pont are common. Normex is made by reacting metaphenylene diamine with isophthalic acid. It is dissolved in solvent and extruded into the



air. The fibre is drawn to develop fibre properties. Kevlar is made from paraphenylene diamine and terephthaloyl chloride. They are first dissolved in a solvent which is then added to a strong mineral acid such as sulphuric acid. Filaments are extruded into a short air space and then into water or dilute sulphuric acid. Drawing is not required to develop fibre properties but fibres are heat treated to increase the molecules of the fibre.

### **Background**

Aramid fibre is an aromatic polyamide, better known by tradenames such as Kevlar (DuPont) and Twaron (Teijin Twaron). As such, many different grades are available, with properties to suit various applications.

It is produced by spinning a solid fibre from solution.

### **Aramid Fiber Characteristics/properties**

- No melting point
- Low flammability
- Good fabric integrity at elevated temperatures
- Para-aramid fibers, which have a slightly different molecular structure, also provide outstanding strength-to-weight properties, high tenacity and high modulus.

### **Some major Aramid Fiber uses**

- Flame-resistant clothing, protective vests and helmets, composites, asbestos replacement, hot air filtration fabrics, tire and mechanical rubber goods reinforcement, ropes and cables, sail cloth, sporting goods.
- Aramid fibers are used in protective garments for firemen, police, and the military. Lighter and tougher than steel, a seven-layer aramid undervest weighing only 2-1/2 pounds can stop a .38-caliber bullet fired from a distance of 10 feet.
- Each year, more ways are found to use these fibers in industrial applications as replacements for steel, fiberglass, asbestos, aluminum, and graphite. Yachts use sails of aramid fibers because of their stability; skis with aramid fiber cores have increased flex life, greater strength, and better performance

## **Applications**

Some applications for aramid fibres are listed below. It is usually used as fibre reinforcement for polymer matrix composites.

- Ballistic protective applications such as bullet proof vests
- Protective apparel such as gloves, motorcycle protective clothing and hunting gaitors, chaps and pants.
- Sails for sailboats, yachts etc
- Belts and hosing for industrial and automotive applications
- Aircraft body parts
- Boat hulls
- Fibre optic and electromechanical cables
- Friction linings such as clutch plates and brake pads
- Gaskets for high temperature and pressure applications
- Adhesives and sealants

## **Key Properties**

Typical properties include:

- Usually yellow in appearance
- Low density
- High strength
- Good impact resistance
- Good abrasion resistance
- Good chemical resistance
- Good resistance to thermal degradation
- Compressive strength similar to E-glass fibres
- Some grades of aramid fibre can degrade when exposed to ultraviolet light